Capstone limitations and revision

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Capstone limitations and revision before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Capstone controlled experiments

Without opening the old lesson, explain one key idea from Capstone controlled experiments. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Documenting a network experiment

Without opening the old lesson, explain one key idea from Documenting a network experiment. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Uncertainty and failure analysis

Without opening the old lesson, explain one key idea from Uncertainty and failure analysis. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Freeze the Week 34 candidate: max(0, a*x), initial a = 1, rate 0.2, two full-batch updates on (1,2), (2,4), no bias. The final weight is 2. The baseline is unchanged at a = 1.

Now open final rows F1=(5,10), F2=(6,12), F3=(-1,-2), F4=(-2,-4). These fictional rows deliberately include a condition absent from the positive **training** and validation **input**s.

Opening them is an evaluation event, not permission to tune on them while retaining an independent-test claim.

Write raw **prediction**s for both procedures before summarising. A low overall error can hide a systematic problem in a condition. These four constructed examples do not estimate prevalence or real-world performance.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Capstone limitations and revision trace and record the first step you can explain without help.

Use W2: Compare with the baseline for Capstone limitations and **revision**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Capstone limitations and revision explanation against the supplied facts before discussing it.

Use the worked case for Capstone limitations and **revision**. Proposed explanation: “Passing a criterion means no failures remain”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Evaluate new evidence independently

A frozen ReLU candidate with a = 3 is compared with baseline a = 1 on final rows (2,6),(4,12), (-1,-3),(-2,-6). Compute both MSE values and candidate MSE on negative **input**s. Write a bounded conclusion.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Trace final predictions

> Trace final predictions. Apply the supplied Capstone limitations and revision facts and keep your reasoning.

For F1–F4 calculate candidate **residual**s and squared errors.

My reasoning and evidence

W2 Compare with the baseline

> Compare with the baseline. Apply the supplied Capstone limitations and revision facts and keep your reasoning.

Compute both overall MSE values and the candidate negative-**input** MSE. Apply the strict final criterion.

My reasoning and evidence

W3 Explain the architecture limit

> Explain the architecture limit. Apply the supplied Capstone limitations and revision facts and keep your reasoning.

Why cannot extra **training** alone make this ReLU **output** negative? Name a proposed change and the **evidence** it would require.

My reasoning and evidence

W4 Evaluate new evidence independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Evaluate new evidence independently. Apply the supplied Capstone limitations and revision facts and keep your reasoning.

A frozen ReLU candidate with a = 3 is compared with baseline a = 1 on final rows (2,6),(4,12), (-1,-3),(-2,-6). Compute both MSE values and candidate MSE on negative **input**s. Write a bounded conclusion.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Label a revision honestly

> Label a revision honestly. Apply the supplied Capstone limitations and revision facts and keep your reasoning.

After inspecting F1–F4, a student changes to **prediction** 2*x and reports “new final MSE 0”. Explain the accurate label and the next evaluation step.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Capstone limitations and revision to Image-pattern Network Investigation, identify a limit and choose the next useful test.

For Image-pattern Network Investigation, use this week’s independent task as a rehearsal for the final **evidence** pack: A frozen ReLU candidate with a = 3 is compared with baseline a = 1 on final rows (2,6),(4,12),(-1,-3),(-2,-6).

Compute both MSE values and candidate MSE on negative **input**s. Write a bounded conclusion. Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

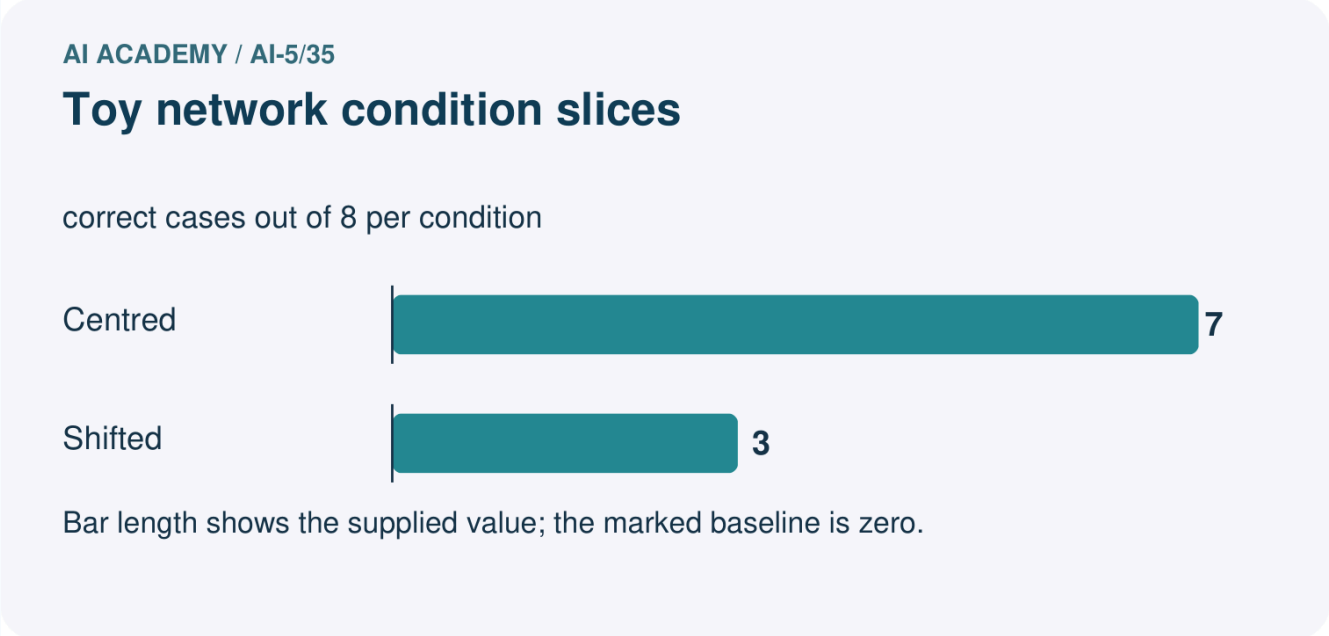
WORKBOOK · 12 OF 15

Pictorial case practice: Capstone limitations and revision

> Revise a limited claim after a slice failure. A toy network works better on centred patterns than on shifted patterns, so the team narrows its initial claim. Explain the labelled step before checking the changed condition.

Revise a limited claim after a slice failure. A toy network works better on centred patterns than on shifted patterns, so the team narrows its initial claim.

Centred slice: 7/8 correct. Shifted slice: 3/8 correct. The network and label rubric are frozen.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/35

Capstone limitations and revision

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute slice rates

Centred is 87.5%; shifted is 37.5%.



2. Revise the claim

Do not describe the network as shift-robust on this evidence.



3. Plan a fair revision

Use separate shifted development patterns, then assess the frozen revision on reserved shifted cases.

CHANGE ONE THING

Train the revision on all eight failed-slice evaluation patterns and rescore them.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Calculate the slice rates and explain the claim **revision**. If the team trains on the shifted evaluation patterns, what kind of **evidence** does their rerun provide?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record.

State exactly what this example supports before linking it to the wider capstone claim.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Image-pattern Network Investigation record.

Project: Image-pattern Network Investigation

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Evidence to keep	My record
This week's input and deciding step	

Evidence to keep	My record
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Revise a limited claim after a slice failure. A toy network works better on centred patterns than on shifted patterns, so the team narrows its initial claim.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Use separate shifted development patterns, then assess the frozen **revision** on reserved shifted cases.

1. Compute slice rates

2. Revise the claim

3. Plan a fair **revision**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Plan a fair **revision** Use separate shifted development patterns, then assess the frozen **revision** on reserved shifted cases.

A limitation guides a better next experiment. **Training** on a failed evaluation example may help development, but that exposed example cannot remain the untouched **evidence** for the **revision**.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Train the **revision** on all eight failed-slice evaluation patterns and rescore them. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Treat the score as exposed-case development, not independent **evidence** of shift robustness.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Treat the score as exposed-case development, not independent **evidence** of shift robustness. The evaluation records now influenced learning and model selection.

Repair a missing building block

Review Capstone controlled experiments (ai-5/34). Goal: Trace matched two-update learning-rate experiments. Short practice: Compute the two correct counts and identify the changed case.

Why would changing B's **training** data weaken the conclusion about its added connection?. Return to this week and explain what you changed.

Review Documenting a network experiment (ai-5/32). Goal: Record a network experiment with enough detail to replay its calculations. Short practice: What additional **evidence** is needed for a final performance claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Uncertainty and failure analysis (ai-5/29). Goal: Distinguish a predicted class, confidence score and observed correctness. Short practice: What is C's positive-class probability and chosen-label confidence?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

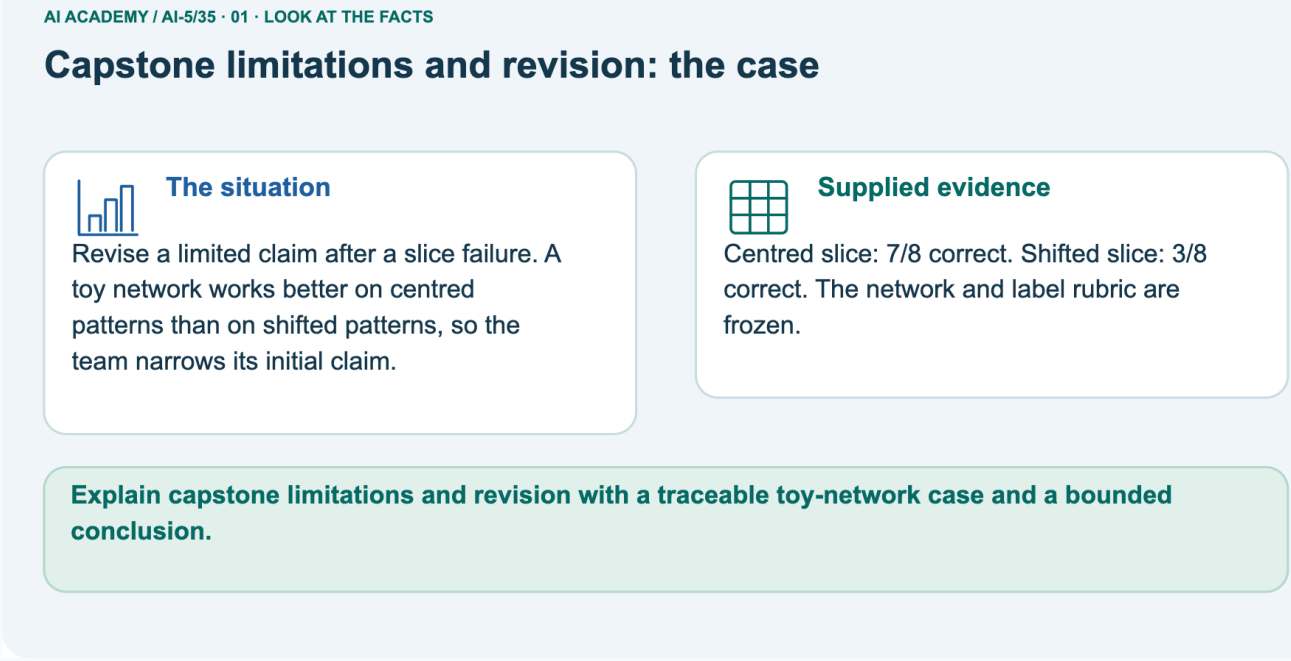
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Capstone limitations and revision picture from its labels, then record a relevant limit.

01. Capstone limitations and revision: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

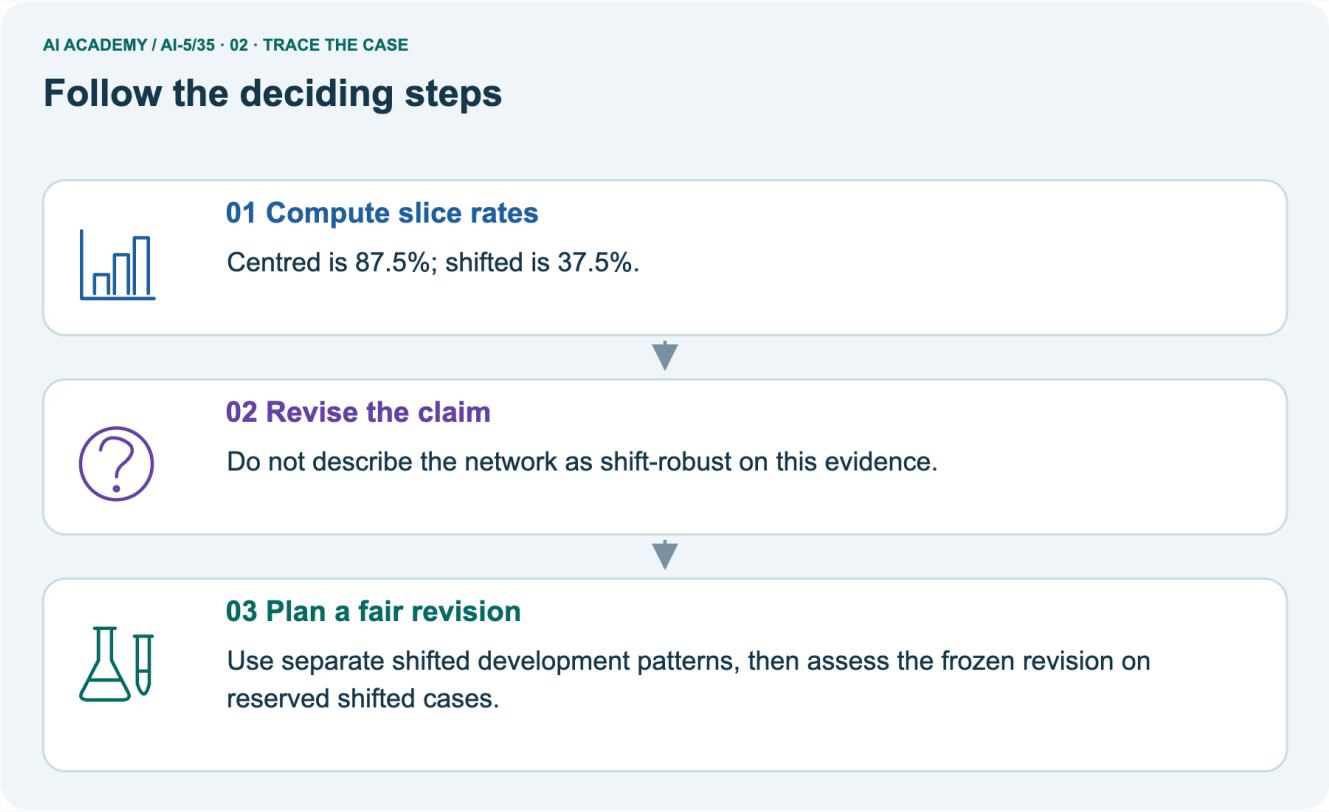
Revise a limited claim after a slice failure. A toy network works better on centred patterns than on shifted patterns, so the team narrows its initial claim. Supplied facts: Centred slice: 7/8 correct.

Shifted slice: 3/8 correct. The network and label rubric are frozen.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Compute slice rates: Centred is 87.5%; shifted is 37.5%. → Revise the claim: Do not describe the network as shift-robust on this **evidence**. → Plan a fair **revision**: Use separate shifted development patterns, then assess the frozen **revision** on reserved shifted cases.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Toy network condition slices



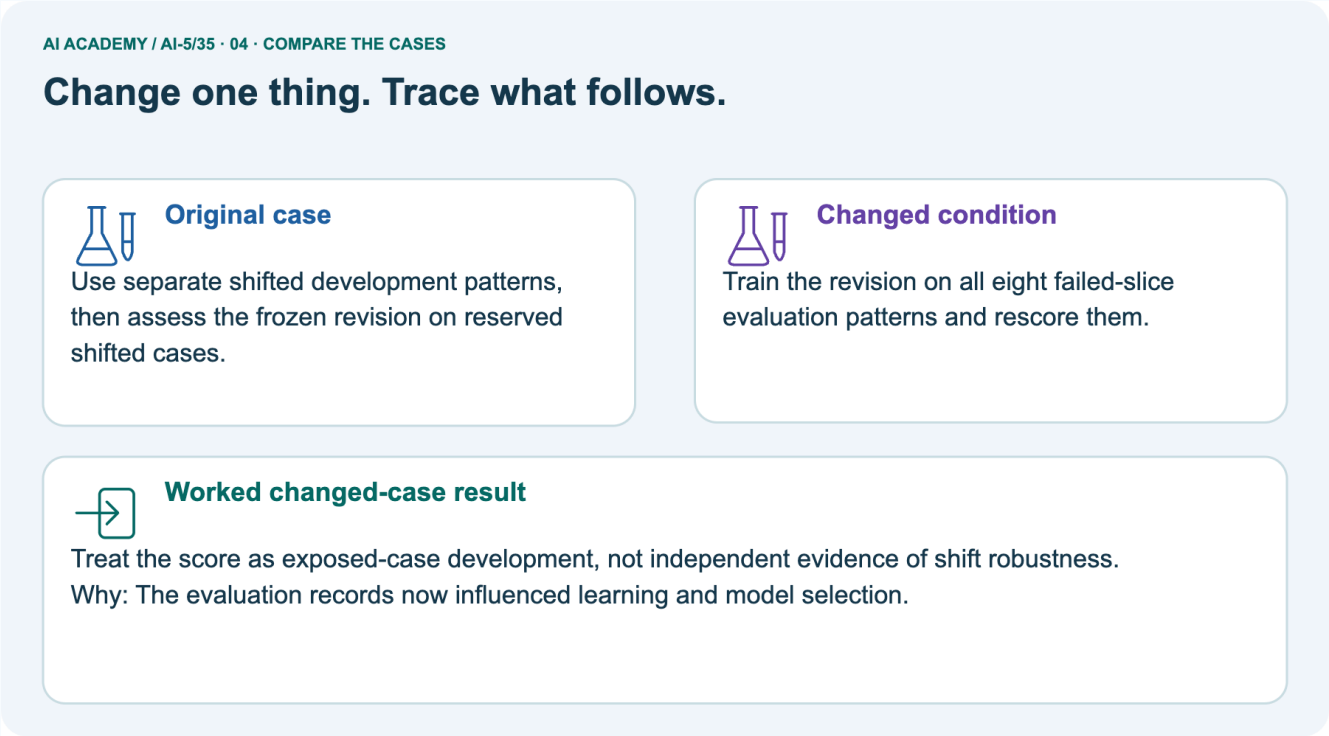
Follow the labelled picture with the matching explanation. Open the vector diagram

Toy network condition slices. Units: correct cases out of 8 per condition. Centred: 7. Shifted: 3.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

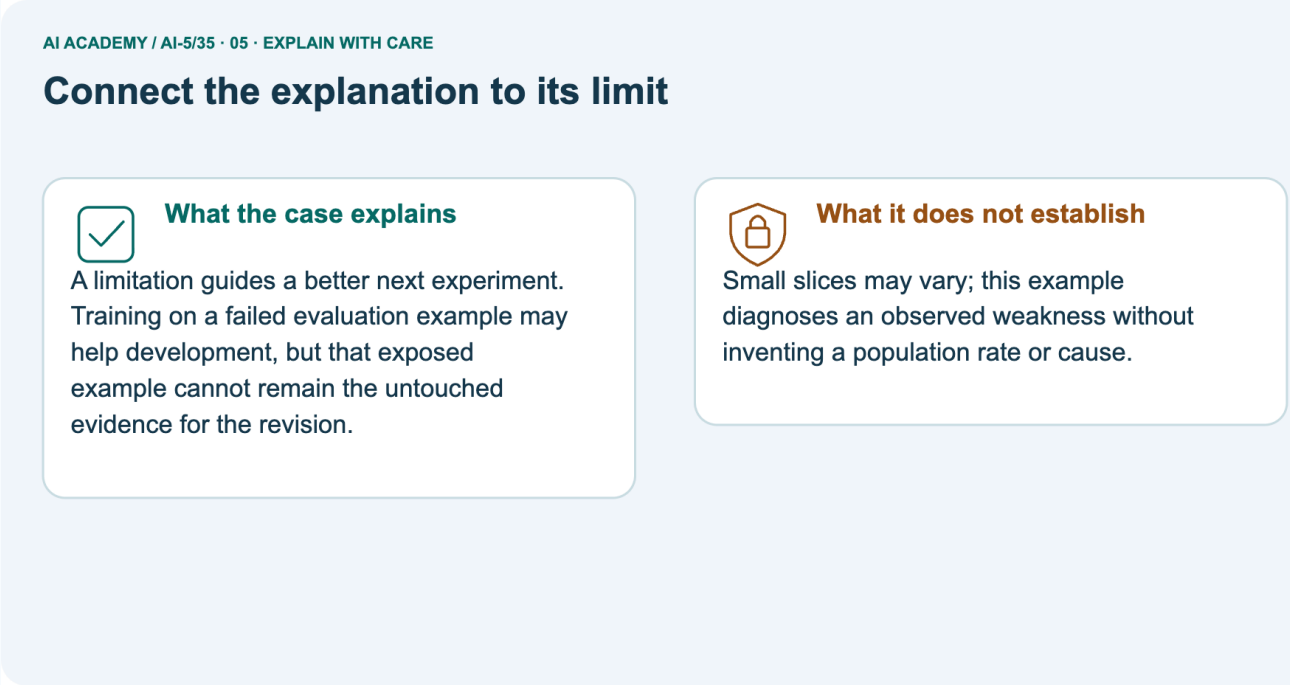
Original case: Use separate shifted development patterns, then assess the frozen **revision** on reserved shifted cases. Changed condition: Train the **revision** on all eight failed-slice evaluation patterns and rescore them.

Expected result: Treat the score as exposed-case development, not independent **evidence** of shift robustness. Reason: The evaluation records now influenced learning and model selection.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

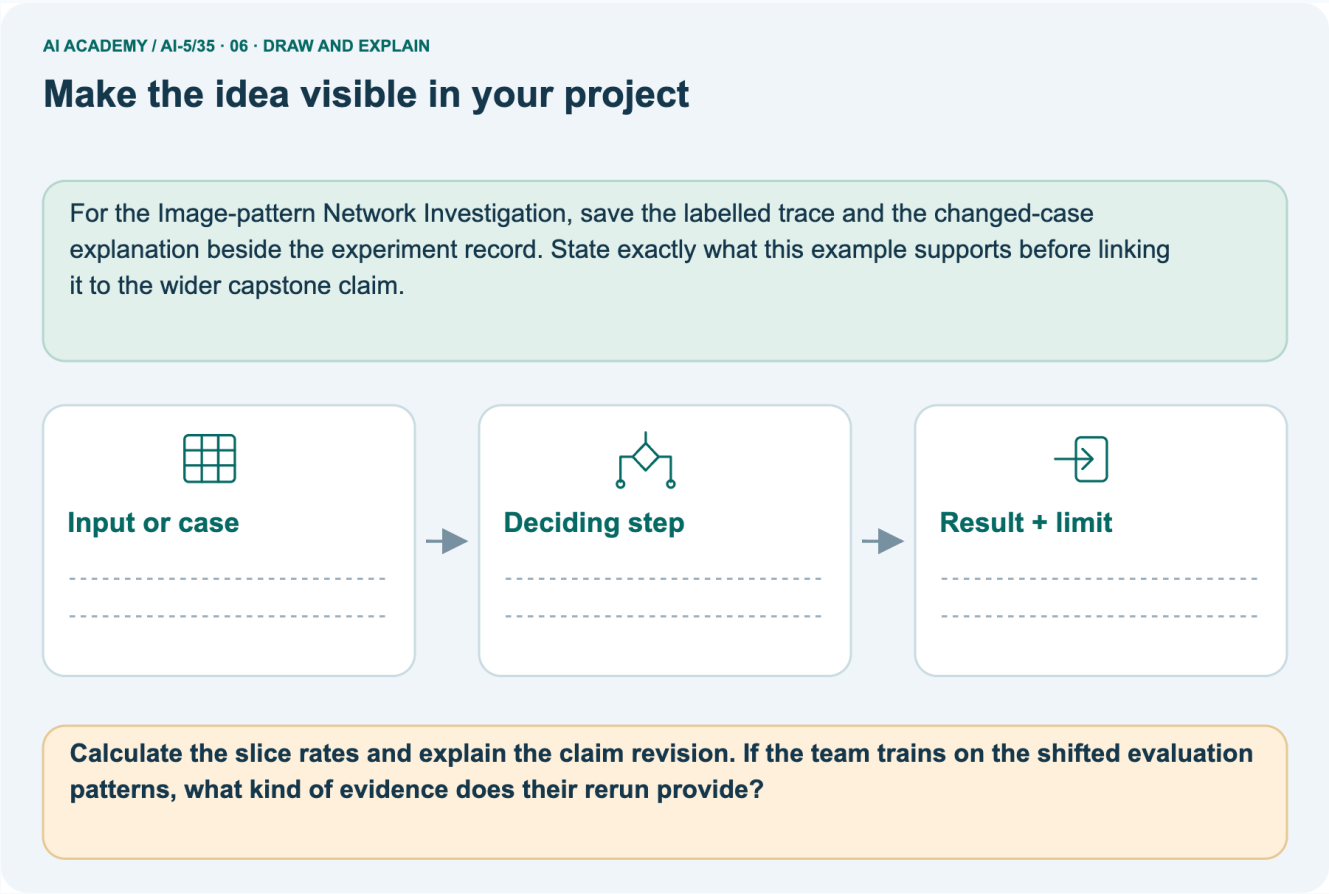
A limitation guides a better next experiment. **Training** on a failed evaluation example may help development, but that exposed example cannot remain the untouched **evidence** for the **revision**.

Limit: Small slices may vary; this example diagnoses an observed weakness without inventing a population rate or cause.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the slice rates and explain the claim **revision**.

If the team trains on the shifted evaluation patterns, what kind of **evidence** does their rerun provide?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.